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Risk of dispersion or aerosol generation and infection transmission with nasopharyngeal and oropharyngeal swabs for detection of COVID-19: a systematic review

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Risk of dispersion or aerosol generation and infection transmission with nasopharyngeal and oropharyngeal swabs for detection of COVID-19: a systematic review.

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Abstract:

Objectives: Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)-related disease, referred to as COVID-19, has emerged as a global pandemic since December 2019. Although transmission is primarily through respiratory droplets and close contact, airborne transmission may be possible in the setting of aerosol generating procedures and treatments. Whether nasopharyngeal and oropharyngeal swabs for COVID-19 generate aerosols remains unclear. This rapid systematic review commissioned by the World Health Organization aimed to determine the risk of aerosol generation and associated transmission with nasopharyngeal and oropharyngeal swabs for detection of COVID-19.

Design: Systematic review.

Data Sources: We searched Ovid MEDLINE and EMBASE from inception until March 25 2020. We also searched the China National Knowledge Infrastructure, Chinese Medical Journal Network, medRxiv and ClinicalTrials.gov as of March 29 2020.

Eligibility Criteria: All comparative and non-comparative studies that evaluated dispersion or aerosolization of viable airborne organisms or transmission of infection associated with nasopharyngeal or oropharyngeal swab testing.

Results: Of 6,640 citations, we retrieved 6 full-text articles. After careful full-text review, no citations were deemed eligible for inclusion.

Conclusions: Aerosol generating procedures may expose healthcare workers and other contacts to pathogens and increase risk of transmission. There is no existing data regarding aerosol generation and risk of transmission with nasopharyngeal and oropharyngeal swabs for the detection of COVID-19. Field experiments to quantify this risk are warranted. Current international guidelines recommend strict droplet and contact precautions during testing.

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Vigilance in adhering to current standards for infection prevention and control measures is suggested.

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Article summary:

- Aerosolization and associated transmission of COVID-19 and other respiratory infections may be possible in the setting of aerosol generating procedures.
- This risk of airborne transmission to uninfected individuals and healthcare workers must be balanced with severe resource constraints in personal protective equipment.
- The risk of aerosol generation associated with nasopharyngeal and oropharyngeal swabs for COVID-19 and other respiratory infections remains unclear, precluding reliable evidence-informed guidance.

Strengths and limitations of this study:

- A comprehensive literature search incorporating English and Chinese-language peer reviewed literature, grey literature and pre-print databases was conducted.
- There was no direct or indirect evidence identified.

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Background:

In late December 2019, a novel coronavirus, subsequently referred to by the World Health Organization (WHO) as *severe acute respiratory syndrome coronavirus 2* (SARS-CoV-2), was identified as the cause of atypical pneumonia cases detected in Wuhan, China.[1] Since then, SARS-CoV-2 related-disease (named COVID-19), has emerged as a global pandemic.[2] As of May 17, SARS-CoV-2 has spread to over 200 countries, infecting 4.7 million individuals, and causing over 313,000 deaths internationally.[3]

COVID-19 is primarily transmitted through mucosal contact with respiratory droplets (5 to 10 μm in diameter) either through close contact (within 1 meter) with an individual who has respiratory symptoms, or through indirect contact with contaminated surfaces.[1,4,5] Airborne transmission may be possible in specific circumstances and settings in which potential aerosol generating procedures or treatments are performed, such as tracheal intubation and bronchoscopy.[5]

On March 29th, 2020, the WHO issued a scientific brief recommending droplet and contact precautions for those people caring for COVID-19 patients and airborne precautions for settings in which aerosol generating procedures and treatments are performed, according to risk assessment.[5] However, crucial knowledge gaps persist regarding which interventions and procedures are associated with clinically significant aerosolization.

Testing for COVID-19 is commonly conducted using nasopharyngeal and oropharyngeal swabs of the posterior wall of the pharynx. The risk of aerosol generation associated with obtaining nasopharyngeal or oropharyngeal swabs in patients infected with COVID-19 remains unclear. The potential to induce coughing may compound this associated risk, the extent to which is unknown.[6]

Given thousands of nasopharyngeal and oropharyngeal swabs are conducted daily internationally, the risk of airborne transmission to otherwise uninfected patients and healthcare

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workers must be balanced with severe resource constraints in personal protective equipment. This dictates an urgent need for an updated evidence synthesis and guidance regarding the risk of aerosolization associated with swab testing. We therefore conducted a rapid systematic review of the literature to determine the risk of aerosol generation and associated transmission associated with nasopharyngeal and oropharyngeal swabs for detection of COVID-19.

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Methods:

Prior to beginning, WHO personnel reviewed and approved internal protocols for this systematic review. Given time constraints of the commissioned review (7 days to completion), the protocol was not registered or published.

Literature search: We conducted a comprehensive search of Ovid MEDLINE and Embase with the assistance of a health information specialist (RC) from inception to March 25, using a combination of subject headings and keywords related to: respiratory tract infections including COVID-19 and other coronaviruses; swabs related to the oral and nasal cavities, nasopharynx and oropharynx; and aerosol generation and infection transmission. We did not limit the search to COVID-19 or coronavirus infection (see Appendix A for the search strategy).

We also applied the same search strategy to the China National Knowledge Infrastructure (CNKI) and Chinese Medical Journal Network (CMJN) (YC). We did not apply any language or quality restrictions. Finally, we searched medRxiv and ClinicalTrials.gov for articles related to COVID-19 or SARS-CoV-2 for pre-prints or grey literature addressing the research question.

Selection criteria: We included all comparative and non-comparative studies that evaluated dispersion or aerosolization of viable airborne organisms or transmission of infection associated with nasopharyngeal or oropharyngeal swab testing. We included studies that evaluated outcomes including detection of viable airborne organisms through microbiological sample analysis or documented transmission of infection associated with swab testing. We initially planned to include studies of hospitalized adult and pediatric patients with microbiologically confirmed COVID-19 in one or more fluid samples, of which at least one was a nasopharyngeal or oropharyngeal swab. In the absence of direct evidence meeting the eligibility criteria above, we broadened our inclusion

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criteria to include all study designs and populations evaluating aerosol generation or dispersion associated with swab testing. This included studies of hospitalized and non-hospitalized patients with or without microbiologically confirmed COVID-19, simulation studies without human participants, and those describing dispersion of non-infectious air particles or liquid droplets.

Study selection: Paired reviewers (SMF, KH, LB, SB) independently screened citations and conducted full-text review of all potentially eligible studies. Reviewers screened the reference lists of review articles to identify additional studies meeting eligibility criteria, and screened pre-prints and grey literature from medRxiv and ClinicalTrials.gov. Disagreements were resolved by a third reviewer (AA) at all stages. Paired reviewers (XX, NY, XL) screened citations identified from the CNKI and CMJN and resolved disagreements by discussion.

Data extraction and quality assessment: Paired reviewers planned to abstract data using a standardized form. We planned to assess risk of bias (RoB) using the modified Cochrane risk of bias tool for randomized controlled trials,[7] the ROBINS-I tool for observational studies,[8] and content expert assessments for experimental or simulation-based studies. We planned to assess the overall certainty of the evidence based on RoB, imprecision, indirectness, inconsistency and publication bias, informed by GRADE guidance.[9,10]

Data analysis: If insufficient data were available for pooled analyses, we planned to summarize results narratively.

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Results:

Of the 6640 citations identified in our search, 4762 were screened for eligibility after duplicates were removed. Of these, 6 were considered potentially eligible requiring full-text review. After careful full-text review, we did not identify any citations evaluating aerosol generation related to nasopharyngeal or oropharyngeal swab testing despite broadening eligibility criteria (Figure 1).

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Discussion

Our systematic review on aerosol generation and associated risk of infection transmission with nasopharyngeal and oropharyngeal swabs in the detection of COVID-19 yielded no relevant studies. Broadening of eligibility criteria to allow for all study designs, including non-comparative and experimental designs and all populations regardless of confirmation of COVID-19 infection, also did not yield any relevant citations.

Airborne transmission is different from droplet transmission, as it refers to the presence of microbes within droplet nuclei. These are generally considered to be particles <5μm in diameter, which can remain in the air for periods of time, and can subsequently be transmitted to others over distances greater than 1 meter.[5] Reduction of respiratory particles to <5 μm involves dehydration of larger droplets and consequent dehydration of any organisms within the droplet. If airborne particles are inhaled, they may rehydrate in the upper airway, increase in size and deposit in the airway. Thus, airborne transmission of viable organisms requires the organism to survive a process of desiccation and aerosolization in sufficient quantity to incite infection. This property is specific to the organism.

Aerosol generating procedures may expose healthcare workers and other contacts to pathogens, and the potential for transmission of respiratory infections. The risk of aerosolization and transmission with swab testing remains unclear, particularly in the context of COVID-19. The most important theoretical risk associated with testing is induction of coughing with possible droplet release and aerosolization. A previous study evaluating voluntary cough-generated aerosolization of influenza virus using a bioaerosol cyclone sampler found that 81% of participants produced detectable viral RNA, with 23% of the RNA contained in particles 1 to 4 micrometres in size and 42% contained in particles less than 1 micrometre.[11] Observational data from Bischoff

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and colleagues showed that patients with confirmed influenza may expose health care workers within 1.83 m to infectious doses of influenza-inoculated droplets.[12] This evidence has guided the WHO to recommend a droplet safety distance of two metres for respiratory illnesses with possible droplet transmission by contact, and appropriate precautions for healthcare workers in close contact with individuals with potential or confirmed infection. COVID-19 has been shown to maintain stability for up to 3 hours in aerosols[13], although this data is based on experimentally induced aerosol-generating procedure with unknown relevance to the clinical setting.

Taken together, recommendations regarding infection prevention and control measures during nasopharyngeal or oropharyngeal swab testing for COVID-19 should be carefully considered in light of the unknown risk of airborne transmission of infection to healthcare workers and other patients. Field experiments with nasopharyngeal and oropharyngeal swab testing in confirmed COVID-19 patients are required to adequately assess this risk using viral samplers, reverse transcriptase polymerase chain reaction testing, and viral cultures. While we cannot estimate the risk of airborne transmission, certain factors may abrogate it: adequate room ventilation, limiting exposure to the patient, high filtration fit-tested respirators (e.g. N95, FFP2) and low viral shedding.[14] However, the risks ought to be weighed against limited resource availability and the need for judicious use of personal protective equipment. One may also consider the yield of various fluid samples, and balance their associated risks of aerosolization with this consideration. The emergence of a diagnostic test with no potential for aerosol generation emerge (e.g. serology) may diminish the enthusiasm for nasopharyngeal and oropharyngeal swabs.

Strengths of this systematic review include a comprehensive literature search incorporating English and Chinese studies, as well as grey literature and pre-prints and consultation with experts in the field. Limitations include a lack of evidence informing the risk of aerosolization with

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nasopharyngeal and oropharyngeal swab testing, including indirect evidence. This highlights an important gap in the existing literature and precludes development of evidence-informed guidance beyond recommendations based on expert opinion.

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Conclusion

There is no evidence regarding the risk of aerosol generation and airborne transmission associated with nasopharyngeal and oropharyngeal swab testing in the context of COVID-19 or related infectious respiratory illnesses. Current international guidelines recommend strict droplet and contact precautions during testing. Given a theoretical risk of cough-related aerosolization during testing, field experiments to quantify the risk of aerosol generation and transmission associated with swab testing are warranted. In the interim, appropriate vigilance in adhering to available standards for infection prevention and control measures is suggested.

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Author Contributions: POV conceived the study. AA and POV organized the study teams and process. AA and RC designed the search strategy. SMF, KH, LB, and SB screened studies for eligibility, and assessed study RoB and certainty of the body of evidence. AA addressed discrepancies in screening. AA wrote the first draft of the manuscript and conducted data analysis. AA, BR, NA, FM, SM, DSH, CG, SM, JVD, KB, JB, FM, DG, DC1, DC2, MH, and POV critically revised the manuscript. AA and POV are the guarantors. All co-authors were involved in final editing of the manuscript, and approved the final draft. The corresponding author attests that all listed authors meet authorship criteria.

Patient and public involvement: No patients were involved in this study, given the severe time constraints of 7 days to completion for this WHO-commissioned rapid review.

Data sharing statement: No additional data available.

Ethics approval statement: Not applicable.

Transparency: The manuscript's guarantors (POV and AA) affirm that the manuscript is an honest, accurate, and transparent account of the recommendation being reported; that no important aspects of the recommendation have been omitted; and that any discrepancies from the recommendation as planned (and, if relevant, registered) have been explained.

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References

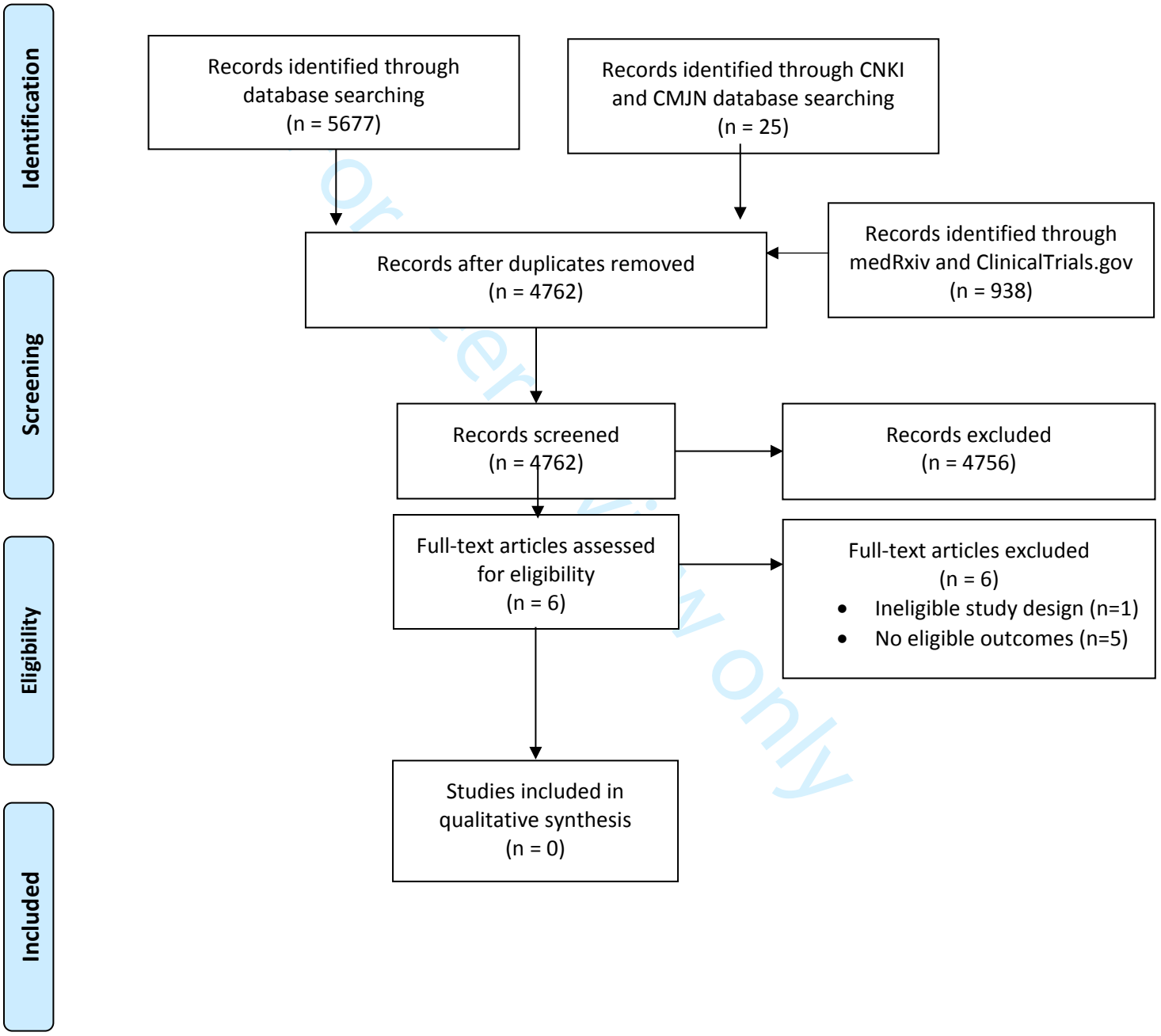
1. Zhu N, Zhang D, Wang W, et al. A novel coronavirus from patients with pneumonia in China, 2019. N Engl J Med. 2020;382:727-33.
2. World Health Organization. Rolling updates on coronavirus disease (COVID-19) [Internet]. 2020 Mar 23 [cited 2020 Mar 24]. Available from: https://www.who.int/emergencies/diseases/novel-coronavirus-2019/events-as-they-happen
3. John Hopkins University of Medicine Coronavirus Resource Centre. Coronavirus COVID-19 global cases by the Center for Systems Science and Engineering (CSSE) at John Hopkins University. 2020 May 17 [cited 2020 May 17]. Available from: https://coronavirus.jhu.edu/map.html
4. Wang W, Xu Y, Gao R, et al. Detection of SARS-CoV-2 in different types of clinical specimens. JAMA. 2020 Mar 11 [Epub ahead of print].
5. World Health Organization. Modes of transmission of virus causing COVID-19: implications for IPC precaution recommendations [Internet]. 2020 Mar 29 [cited 2020 Mar 29]. Available from: https://www.who.int/news-room/commentaries/detail/modes-of-transmission-of-virus-causing-covid-19-implications-for-ipc-precaution-recommendations
6. World Health Organization. Q&A on infection prevention and control for health care workers caring for patients with suspected or confirmed 2019-nCoV [Internet]. 2020 Mar 01 [cited 2020 Mar 31]. Available from: https://www.who.int/news-room/q-a-detail/q-a-on-infection-prevention-and-control-for-health-care-workers-caring-for-patients-with-suspected-or-confirmed-2019-ncov.

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7. Higgins JPT, Altman DG, Gotzsche PC, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. BMJ. 2011;343:d5928.
8. Sterne JA, Hernán MA, Reeves BC, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. BMJ. 2016;355:i4919.
9. Guyatt GH, Oxman AD, Vist GE, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. BMJ (Clinical Research ed). 2008;336(7650):924-6.
10. Morgan RL, Thayer KA, Bero L, et al. GRADE: Assessing the quality of evidence in environmental and occupational health. Environ Int. 2016;92-93:611-6.
11. Lindsley WG, Blachere FM, Thewlis RE, et al. Measurements of airborne influenza virus in aerosol particles from human coughs. PLoS One. 2010;5(11):e15100.
12. Bischoff WE, Swett K, Leng I, Peters TR. Exposure to influenza virus aerosols during routine patient care. J Infect Dis. 2013;207(7):1037-46.
13. van Doremalen N, Bushmaker T, Morris DH, et al. Aerosol and surface stability of SARS-CoV-2 as compared with SARS-CoV-1. N Engl J Med. 2020 [Epub ahead of print].
14. Fennelly KP, Nardell EA. The relative efficacy of respirators and room ventilation in preventing occupational tuberculosis. Infect Cont Hosp Ep. 1998;19:754-9.

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Figure 1: PRISMA flow diagram for review on aerosol generation associated with nasopharyngeal and oropharyngeal swabs.



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Appendix A: Search strategy for OVID Medline and EMBASE.

MEDLINE
Database: OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:
---
1 exp Respiratory Tract Infections/ (352559)
2 (respiratory adj3 infect*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (75025)
3 (influenza* or flu).tw. (121359)
4 (h1n1 or h5n1 or h3n2).tw. (25097)
5 exp Coronavirus/ (11467)
6 exp Coronavirus Infections/ (9776)
7 (coronavir* or coronovir* or SARS or MERS or MERS-COV or SARS-COV or SARS-COV-2 or COV or NCOV or 2019nCOV or 2019-nCOV or COVID-19).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (24352)
8 or/1-7 (469619)
9 ((throat or pharynx or pharyng* or oral or oropharyng* or nasal or nares or nasopharyng*) adj3 (swab* or detect* or test or sample* or specimen* or diagnos* or assay)).mp. (49581)
10 8 and 9 (8147)
11 exp Disease Transmission, Infectious/ (67129)
12 (contamina* or dispers* or droplet* or partic* or aerosol* or environment* or transmiss* or transmit or safety* or commun* or contagi*).mp. (5570978)
13 11 or 12 (5584372)
14 10 and 13 (2488)

EMBASE
Database: Embase <1974 to 2020 March 24>
Search Strategy:
---
1 exp respiratory tract infection/ (385263)
2 (respiratory adj3 infect*).mp. (127592)
3 (influenza* or flu).tw. (139944)
4 (h1n1 or h5n1 or h3n2).tw. (30836)
5 exp coronavirinae/ (11209)
6 exp Coronavirus infection/ (11152)
7 (coronavir* or coronovir* or SARS or MERS or MERS-COV or SARS-COV or SARS-COV-2 or COV or NCOV or 2019nCOV or 2019-nCOV or COVID-19).mp. (28959)
8 or/1-7 (491703)
9 ((throat or pharynx or pharyng* or oral or oropharyng* or nasal or nares or nasopharyng*) adj3 (swab* or detect* or test or sample* or specimen* or diagnos* or assay)).mp. (80034)

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10 8 and 9 (9252)
11 disease transmission/ (96856)
12 (contamina* or dispers* or droplet* or partic* or aerosol* or environment* or transmiss* or transmit or safety* or commun* or contagi*).mp. (7223561)
13 11 or 12 (7223561)
14 10 and 13 (3189)

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5-6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5-6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	7
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	N/A
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	N/A
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	N/A
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	9
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	9-11
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	12

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Risk of dispersion or aerosol generation and infection transmission with nasopharyngeal and oropharyngeal swabs for detection of COVID-19: a systematic review

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Risk of dispersion or aerosol generation and infection transmission with nasopharyngeal and oropharyngeal swabs for detection of COVID-19: a systematic review.

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Abstract:

Objectives: Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)-related disease, referred to as COVID-19, has emerged as a global pandemic since December 2019. While there is growing recognition regarding possible airborne transmission, particularly in the setting of aerosol-generating procedures and treatments, whether nasopharyngeal and oropharyngeal swabs for SARS-CoV-2 generate aerosols remains unclear.

Design: Systematic review.

Data Sources: We searched Ovid MEDLINE and EMBASE up to November 3 2020. We also searched the China National Knowledge Infrastructure, Chinese Medical Journal Network, medRxiv and ClinicalTrials.gov up to March 29 2020.

Eligibility Criteria: All comparative and non-comparative studies that evaluated dispersion or aerosolization of viable airborne organisms, or transmission of infection associated with nasopharyngeal or oropharyngeal swab testing.

Results: Of 7,702 citations, only one study was deemed eligible. Using a dedicated sampling room with negative pressure isolation room, personal protective equipment including N95 or higher masks, strict sterilization protocols, structured training with standardized collection methods, and a structured collection and delivery system, a tertiary care hospital proved a 0% healthcare worker infection rate among 8 nurses conducting over 11,000 nasopharyngeal swabs. No studies examining transmissibility with other safety protocols, nor any studies quantifying the risk of aerosol generation with nasopharyngeal or oropharyngeal swabs for detection of SARS-CoV-2 were identified.

Conclusions: There is limited to no published data regarding aerosol generation and risk of transmission with nasopharyngeal and oropharyngeal swabs for the detection of SARS-CoV-2.

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Field experiments to quantify this risk are warranted. Vigilance in adhering to current standards for infection control is suggested.

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Article summary:

Strengths and limitations of this study:

- A comprehensive literature search incorporating English and Chinese-language peer reviewed literature, grey literature and pre-print databases was conducted.
- The literature search included in-patient and outpatient testing settings.
- The literature search accounted for available direct and indirect evidence.
- Findings of this systematic review were limited to one study evaluating transmission risk to healthcare personnel conducting nasopharyngeal swabs; no studies directly quantified the risk of airborne transmission.
- The findings of the systematic review preclude substantive evidence-informed guidance, and warrant vigilance in adhering to current standards for infection prevention and control measures.

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Background:

In late December 2019, a novel coronavirus, subsequently referred to by the World Health Organization (WHO) as *severe acute respiratory syndrome coronavirus 2* (SARS-CoV-2), was identified as the cause of atypical pneumonia cases detected in Wuhan, China (1). Since then, SARS-CoV-2 related-disease (named COVID-19), has emerged as a global pandemic (2). In one year, SARS-CoV-2 has spread to over 200 countries, infecting over 90 million individuals, and causing over 2 million deaths internationally (3).

International infection control guidance initially stated that viral transmission is likely primarily through direct, indirect or close contact with infected salivary and respiratory droplets (5 to 10 μm in diameter) (4). More recently, the WHO and the Centers for Disease Control and Prevention have recognized that there is growing evidence regarding SARS-CoV-2 transmission from airborne exposure, particularly in poorly ventilated enclosed settings with extended exposures, and in the setting of aerosol-generating procedures in healthcare settings (5-8).

Testing for COVID-19 is commonly conducted using nasopharyngeal and oropharyngeal swabs of the posterior wall of the pharynx. The risk of aerosol generation associated with obtaining nasopharyngeal or oropharyngeal swabs in patients infected with COVID-19 remains unclear. The potential to induce coughing may compound this associated risk, the extent to which is unknown.

Given thousands of nasopharyngeal and oropharyngeal swabs are conducted daily internationally, the risk of potential airborne transmission to otherwise uninfected patients and healthcare workers must be balanced with severe resource constraints in personal protective equipment. This dictates an urgent need for an updated evidence synthesis and guidance regarding the risk of aerosolization associated with swab testing. We therefore conducted a systematic review of the literature to determine the risk of aerosol generation and associated transmission associated

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with nasopharyngeal and oropharyngeal swabs for detection of SARS-CoV-2.

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Methods:

Prior to initiating this systematic review, WHO personnel reviewed and approved internal protocols for this systematic review. Given time constraints of the initial commissioned review for the WHO in May 2020 (7 days to completion), the protocol was not publicly registered or published.

Literature search: We conducted a comprehensive search of Ovid MEDLINE and Embase with the assistance of a health information specialist (RC) from inception to November 3 2020 (updated), using a combination of subject headings and keywords related to: respiratory tract infections including COVID-19 and other coronaviruses; swabs related to the oral and nasal cavities, nasopharynx and oropharynx; and aerosol generation and infection transmission. We did not limit the search to COVID-19 or coronavirus infection (see Appendix A for the search strategy).

We also applied the same search strategy as of March 25 2020 to the China National Knowledge Infrastructure (CNKI) and Chinese Medical Journal Network (CMJN) (YC). We did not apply any language or quality restrictions. Finally, we searched medRxiv and ClinicalTrials.gov for articles related to COVID-19 or SARS-CoV-2 for pre-prints or grey literature addressing the research question.

Selection criteria: We included all comparative and non-comparative studies that evaluated dispersion or aerosolization of viable airborne organisms or transmission of infection associated with nasopharyngeal or oropharyngeal swab testing. We included studies that evaluated outcomes including detection of viable airborne organisms through microbiological sample analysis or documented transmission of infection associated with swab testing. We initially planned to include

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studies of hospitalized adult and pediatric patients with microbiologically confirmed COVID-19 in one or more fluid samples, of which at least one was a nasopharyngeal or oropharyngeal swab. In the absence of direct evidence meeting the eligibility criteria above, we broadened our inclusion criteria to include all study designs and populations evaluating aerosol generation or dispersion associated with swab testing. This included studies of hospitalized and non-hospitalized patients with or without microbiologically confirmed COVID-19, simulation studies without human participants, and those describing dispersion of non-infectious air particles or liquid droplets.

Study selection: Paired reviewers (SMF, KH, LB, SB) independently screened citations and conducted full-text review of all potentially eligible studies. Reviewers screened the reference lists of review articles to identify additional studies meeting eligibility criteria, and screened pre-prints and grey literature from medRxiv and ClinicalTrials.gov. Disagreements were resolved by a third reviewer (AA) at all stages. Paired reviewers (XX, NY, XL) screened citations identified from the CNKI and CMJN and resolved disagreements by discussion.

Data extraction and quality assessment: Paired reviewers planned to abstract data using a standardized form. We planned to assess risk of bias (RoB) using the modified Cochrane risk of bias tool for randomized controlled trials (9), the ROBINS-I tool for observational studies (10), and content expert assessments for experimental or simulation-based studies. We planned to assess the overall certainty of the evidence based on RoB, imprecision, indirectness, inconsistency and publication bias, informed by GRADE guidance (11, 12).

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Data analysis: If insufficient data were available for pooled analyses, we planned to summarize results narratively.

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Results:

Of the 7,702 citations identified in our search, 5,477 were screened for eligibility after duplicates were removed. Of these, 9 were considered potentially eligible requiring full-text review. After careful full-text review, we identified one observational study examining transmission rates associated with a safety protocol for nasopharyngeal swabs for detection of SARS-CoV-2 (13). This study implemented and evaluated a safety protocol for nasopharyngeal swab sampling in a tertiary care hospital in Wuhan, China. The protocol involved establishment of a special sampling room with negative pressure ventilation, training of nursing staff conducting swab tests, strict sterilization protocols, personal protective equipment including N95 or higher-level masks with standard droplet and contact precautions, and standardized swab collection and transportation methods. Based on preliminary evaluation among eight nurses conducting over 11,000 nasopharyngeal swabs, a 0% nursing staff infection rate was reported (13).

We did not apply the ROBINS-I tool, given the study was non-comparative. Overall certainty of evidence was very low, with very serious concerns regarding risk of bias, imprecision and publication bias.

No studies evaluating dispersion and aerosol generation associated with nasopharyngeal or oropharyngeal swabs, nor any studies comparing transmission using different safety protocols, were found despite broadening eligibility criteria (Figure 1).

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Discussion

This systematic review on aerosol generation and associated risk of infection transmission with nasopharyngeal and oropharyngeal swabs in the detection of SARS-CoV-2 yielded limited to no evidence. Only one small study evaluating a preliminary safety protocol for nasopharyngeal swabs was found, which showed that standardized protocols and airborne precautions were associated with no infections among nursing staff conducting testing (13). The safety protocol implemented, which included a negative pressure room and N95 or higher masks, is not widely implemented in current practice; no data was provided regarding risk of transmission with nasopharyngeal swabs under other settings. Broadening of eligibility criteria to allow for all study designs, including non-comparative and experimental designs and all populations regardless of COVID-19 infection, also did not yield any relevant citations.

Airborne transmission is different from droplet transmission, as it refers to the presence of microbes within droplet nuclei. These are generally considered to be particles <5μm in diameter, which can remain in the air for periods of time, and can subsequently be transmitted to others over distances greater than 1 meter (6). Reduction of respiratory particles to <5 μm involves dehydration of larger droplets and consequent dehydration of any organisms within the droplet. If airborne particles are inhaled, they may rehydrate in the upper airway, increase in size and deposit in the airway. Thus, airborne transmission of viable organisms requires the organism to survive a process of desiccation and aerosolization in sufficient quantity to incite infection. This property is specific to the organism (14, 15).

Aerosol generating procedures may expose healthcare workers and other contacts to pathogens, and the potential for transmission of respiratory infections. The risk of aerosolization and transmission with swab testing remains unclear, particularly in the context of COVID-19. The

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most important theoretical risk associated with testing is induction of coughing with possible droplet release and aerosolization. A previous study evaluating voluntary cough-generated aerosolization of influenza virus using a bioaerosol cyclone sampler found that 81% of participants produced detectable viral RNA, with 23% of the RNA contained in particles 1 to 4 micrometres in size and 42% contained in particles less than 1 micrometre (16).

Specific to COVID-19, experimental studies have shown aerosols with SARS-CoV-2 virus RNA in air samples for up to 3 to 16 hours (17, 18); however, studies from healthcare settings with symptomatic COVID-19 patients have yielded mixed results regarding identification of SARS-CoV-2 in air samples, and none have identified viable virus (6).

Initial guidance from the WHO on March 29, 2020 recommended droplet and contact precautions for those people caring for COVID-19 patients, and airborne precautions for settings in which aerosol generating procedures and treatments are performed (4). Subsequent guidance on December 1 2020 recommended, in addition to standard droplet and contact precautions, that medical masks in the absence of aerosol generating procedures, and respirators, N95s, or equivalent level masks be worn in care settings where such procedures are performed. Aerosol generating procedures were defined as "tracheal intubation, non-invasive ventilation, tracheotomy, cardiopulmonary resuscitation, manual ventilation before intubation, bronchoscopy, sputum induction using nebulized hypertonic saline, and dentistry and autopsy procedures" (7). Guidance specific to nasopharyngeal and oropharyngeal swab testing were not explicitly provided.

Recent evidence regarding salivary nucleic acid amplification testing has supported comparable diagnostic accuracy, with a pooled sensitivity of 83.2% (95% confidence intervals 74.7 to 91.4%) and specificity of 99.2% (98.2 to 99.8%), compared to nasopharyngeal swab testing (84.8% [76.8% to 92.4%] and 98.9% [97.4% to 99.8%]), respectively (19). While demonstrating

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comparable yield, salivary testing was associated with lower costs (20). Coupled with increased ease of testing, reduced invasiveness and likely reduced occupational exposure risk, salivary testing may be a viable alternative to nasopharyngeal or oropharyngeal swab testing.

Taken together, recommendations regarding infection prevention and control measures during nasopharyngeal or oropharyngeal swab testing for SARS-CoV-2 should be carefully considered in light of the unknown risk of airborne transmission of infection to healthcare workers and other patients. Field experiments with nasopharyngeal and oropharyngeal swab testing in confirmed COVID-19 patients are required to adequately assess this risk using viral samplers, reverse transcriptase polymerase chain reaction testing, and viral cultures. While we cannot estimate the risk of airborne transmission, certain factors may abrogate it: adequate room ventilation, limiting exposure to the patient, high filtration fit-tested respirators (e.g. N95, FFP2) and low viral shedding (13, 21). However, the risks ought to be weighed against limited resource availability and the need for judicious use of personal protective equipment. One may also consider the yield of various fluid samples and balance their associated risks of aerosolization with this consideration. The emergence of diagnostic tests such as salivary testing with less potential for aerosol generation may diminish the enthusiasm for nasopharyngeal and oropharyngeal swabs.

Strengths of this systematic review include a comprehensive literature search incorporating English and Chinese studies, as well as grey literature and pre-prints and consultation with experts in the field. Limitations include a lack of published evidence informing the risk of aerosolization with nasopharyngeal and oropharyngeal swab testing, including indirect evidence. This highlights an important gap in the existing literature and precludes development of evidence-informed guidance beyond recommendations based on expert opinion.

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Conclusion

Our systematic review revealed limited to no data specifically addressing the risk of aerosol generation and airborne transmission associated with nasopharyngeal and oropharyngeal swab testing in the context of COVID-19 or related infectious respiratory illnesses. Current international guidelines recommend droplet and contact precautions during testing. Given a theoretical risk of cough-related aerosolization during testing, field experiments to quantify the risk of aerosol generation and transmission associated with swab testing are warranted. In the interim, appropriate vigilance in adhering to available standards for infection prevention and control measures is suggested. Alternative testing modalities, including salivary testing, may warrant consideration in place of nasopharyngeal and oropharyngeal testing.

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Author Contributions: Per O. Vandvik conceived the study. Arnav Agarwal and Per O. Vandvik organized the study teams and process. Arnav Agarwal and Rachel Couban designed the search strategy. Shannon M. Fernando, Kimia Honarmand, Layla Bakaa, and Sonia Brar screened studies for eligibility, and assessed study risk of bias and certainty of the body of evidence. Arnav Agarwal addressed discrepancies in screening. Arnav Agarwal wrote the first draft of the manuscript and conducted data analysis. Arnav Agarwal, Bram Rochwerg, Neill K. Adhikari, Francois Lamontagne, Srinivas Murthy, David S.C. Hui, Charles D. Gomersall, Samira Mubareka, Janet V. Diaz, Karen E.A. Burns, John Basmaji, Fiona Muttalib, David Granton, Dipayan Chaudhuri, Devin Chetan, Malini Hu, and Per O. Vandvik critically revised the manuscript. Arnav Agarwal and Per O. Vandvik are the guarantors. All co-authors were involved in final editing of the manuscript, and approved the final draft. The corresponding author attests that all listed authors meet authorship criteria.

Patient and public involvement: No patients were involved in this study, given the severe time constraints of 7 days to completion for the initial WHO-commissioned rapid review.

Data sharing statement: No additional data available.

Ethics approval statement: Not applicable.

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Transparency: The manuscript's guarantors (POV and AA) affirm that the manuscript is an honest, accurate, and transparent account of the recommendation being reported; that no important aspects of the recommendation have been omitted; and that any discrepancies from the recommendation as planned (and, if relevant, registered) have been explained.

Competing Interests: No competing interests declared.

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References

1. Zhu N, Zhang D, Wang W, Li X, Yang B, Song J, et al. A novel coronavirus from patients with pneumonia in China, 2019. N Engl J Med. 2020;382(8):727-33.

2. World Health Organization. Rolling updates on coronavirus disease (COVID-19). Geneva, Switzerland: World Health Organization; 2020.

3. John Hopkins Coronavirus Resource Centre. Coronavirus COVID-19 global cases by the Center for Systems Science and Engineering (CSSE) at John Hopkins University. Baltimore, Maryland: John Hopkins University; 2020.

4. World Health Organization. Modes of transmission of virus causing COVID-19: implications for IPC precaution recommendations. Geneva, Switzerland: World Health Organization; 2020.

5. Centers for Disease Control and Prevention. Scientific brief: SARS-CoV-2 and potential airborne transmission Atlanta, Georgia: U.S. Department of Health & Human Services; 2020.

6. World Health Organization. Transmission of SARS-CoV-2: implications for infection prevention precautions. Geneva, Switzerland: World Health Organization; 2020.

7. World Health Organization. Mask use in the context of COVID-19. Geneva, Switzerland: World Health Organization; 2020.

8. World Health Organization. Q&A on infection prevention and control for health care workers caring for patients with suspected or confirmed 2019-nCoV. Geneva, Switzerland: World Health Organization; 2020.

9. Higgins JP, Altman DG, Gøtzsche PC, Jüni P, Moher D, Oxman AD, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. BMJ. 2011;343:d5928.

For peer review only

10. Sterne JA, Hernán MA, Reeves BC, Savović J, Berkman ND, Viswanathan M, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. BMJ. 2016;355:i4919.

11. Guyatt GH, Oxman AD, Vist GE, Kunz R, Falck-Ytter Y, Alonso-Coello P, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. BMJ. 2008;336(7650):924-6.

12. Morgan RL, Thayer KA, Bero L, Bruce N, Falck-Ytter Y, Ghersi D, et al. GRADE: Assessing the quality of evidence in environmental and occupational health. Environ Int. 2016;92-93:611-6.

13. Qian Y, Zeng T, Wang H, Xu M, Chen J, Hu N, et al. Safety management of nasopharyngeal specimen collection from suspected cases of coronavirus disease 2019. Int J Nurs Sci. 2020;7(2):153-6.

14. Pepper IL, Gerba CP. Aeromicrobiology. In: Pepper I, Gerba C, Gentry T, editors. Environmental Microbiology 3rd ed ed. Tucson, Arizona: Elsevier Inc; 2015. p. 89-110.

15. Cox CS. Airborne bacteria and viruses. Sci Prog. 1989;73(292 Pt 4):469-99.

16. Lindsley WG, Blachere FM, Thewlis RE, Vishnu A, Davis KA, Cao G, et al. Measurements of airborne influenza virus in aerosol particles from human coughs. PLoS One. 2010;5(11):e15100.

17. van Doremalen N, Bushmaker T, Morris DH, Holbrook MG, Gamble A, Williamson BN, et al. Aerosol and surface stability of SARS-CoV-2 as compared with SARS-CoV-1. N Engl J Med. 2020;382(16):1564-7.

For peer review only

18. Fears AC, Klimstra WB, Duprex P, Hartman A, Weaver SC, Plante KS, et al. Persistence of severe acute respiratory syndrome coronavirus 2 in aerosol suspensions. Emerg Infect Dis. 2020;26(9).

19. Butler-Laporte G, Lawandi A, Schiller I, Yao MC, Dendukuri N, McDonald EG, et al. Comparison of saliva and nasopharyngeal swab nucleic acid amplification testing for detection of SARS-CoV-2: a systematic review and meta-analysis. JAMA Intern Med. 2021 [Epub ahead of print].

20. Bastos ML, Perlman-Arrow S, Menzies D, Campbell JR. The sensitivity and costs of testing for SARS-CoV-2 infection with saliva versus nasopharyngeal swabs: a systematic review and meta-analysis. Ann Intern Med. 2021 [Epub ahead of print].

21. Fennelly KP, Nardell EA. The relative efficacy of respirators and room ventilation in preventing occupational tuberculosis. Infect Control Hosp Epidemiol. 1998;19(10):754-9.

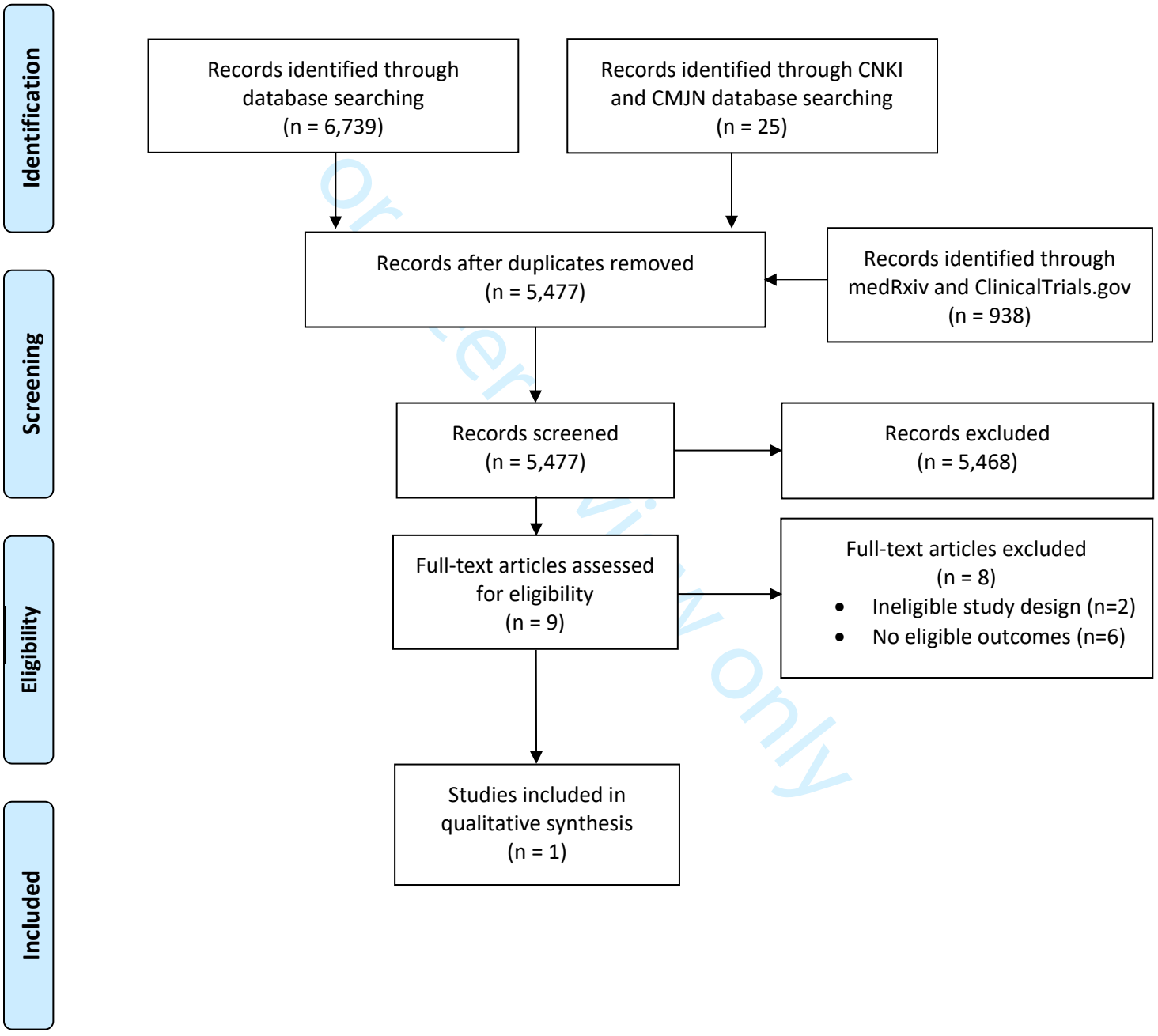
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Figure legends

Figure 1: PRISMA flow diagram for review on aerosol generation associated with nasopharyngeal and oropharyngeal swabs.

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Figure 1: PRISMA flow diagram for review on aerosol generation associated with nasopharyngeal and oropharyngeal swabs.



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Appendix A: Search strategy for OVID Medline and EMBASE.

MEDLINE
Database: OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present
Search Strategy:
---
1 exp Respiratory Tract Infections/ (352559)
2 (respiratory adj3 infect*).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (75025)
3 (influenza* or flu).tw. (121359)
4 (h1n1 or h5n1 or h3n2).tw. (25097)
5 exp Coronavirus/ (11467)
6 exp Coronavirus Infections/ (9776)
7 (coronavir* or coronovir* or SARS or MERS or MERS-COV or SARS-COV or SARS-COV-2 or COV or NCOV or 2019nCOV or 2019-nCOV or COVID-19).mp. [mp=title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms] (24352)
8 or/1-7 (469619)
9 ((throat or pharynx or pharyng* or oral or oropharyng* or nasal or nares or nasopharyng*) adj3 (swab* or detect* or test or sample* or specimen* or diagnos* or assay)).mp. (49581)
10 8 and 9 (8147)
11 exp Disease Transmission, Infectious/ (67129)
12 (contamina* or dispers* or droplet* or partic* or aerosol* or environment* or transmiss* or transmit or safety* or commun* or contagi*).mp. (5570978)
13 11 or 12 (5584372)
14 10 and 13 (2488)

EMBASE
Database: Embase <1974 to 2020 March 24>
Search Strategy:
---
1 exp respiratory tract infection/ (385263)
2 (respiratory adj3 infect*).mp. (127592)
3 (influenza* or flu).tw. (139944)
4 (h1n1 or h5n1 or h3n2).tw. (30836)
5 exp coronavirinae/ (11209)
6 exp Coronavirus infection/ (11152)
7 (coronavir* or coronovir* or SARS or MERS or MERS-COV or SARS-COV or SARS-COV-2 or COV or NCOV or 2019nCOV or 2019-nCOV or COVID-19).mp. (28959)
8 or/1-7 (491703)
9 ((throat or pharynx or pharyng* or oral or oropharyng* or nasal or nares or nasopharyng*) adj3 (swab* or detect* or test or sample* or specimen* or diagnos* or assay)).mp. (80034)

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10 8 and 9 (9252)
11 disease transmission/ (96856)
12 (contamina* or dispers* or droplet* or partic* or aerosol* or environment* or transmiss* or transmit or safety* or commun* or contagi*).mp. (7223561)
13 11 or 12 (7223561)
14 10 and 13 (3189)

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5-6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5-6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	7
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7-8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	8
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	9

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	10
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	10
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	10
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	11
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	13
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	14
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	15

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